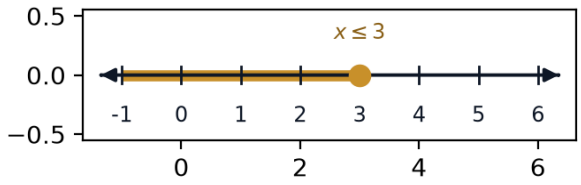
# Linear Functions & Inequalities

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01 • 20 FRQs + 25 MCQs • Difficulty Progression • Cross-Lesson Synthesis

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • [FRQ] • Target: Solving One-Variable Linear Inequalities

Solve each linear inequality for $x$ and express the solution set in interval notation:

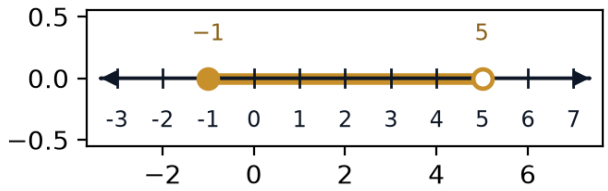



**a** $7 - 4(2x - 3) \geq 3x - 14$

**b** $-9 < \frac{5 - 3x}{2} \leq 6$

**2** [Foundational] • [FRQ] • Target: Compound Inequalities & Number Line Graphs

Solve the compound inequality $-5 \leq 2x - 3 < 7$ and verify that the number line below accurately represents its solution set.




**3** [Foundational] • [FRQ] • Target: Solving Systems by Substitution

Solve the linear system by substitution and state the value of $x + y$: $y = 3x - 7$ and $4x + 3y = 31$.

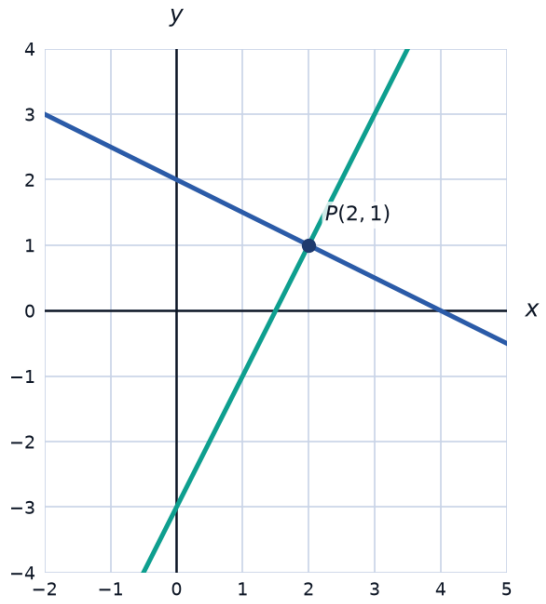
**4** [Foundational] • [FRQ] • Target: Solving Systems by Elimination

Solve the linear system by elimination: $5x - 2y = 26$ and $3x + 4y = 10$.

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**5** [Foundational] • [FRQ] • Target: Graphical Intersection of Linear Systems

The graphs of $y = 2x - 3$ and $y = -\frac{1}{2}x + 2$ are shown in the $xy$-plane. Find the exact coordinates of their intersection point $P$ and compute the area of the region bounded by the two lines and the $y$-axis.




**6** [Foundational] • [FRQ] • Target: Two-Variable System Word Problems

A university lecture hall has $320$ seats divided between standard seats ($25$ dollars each) and VIP balcony seats ($45$ dollars each). If a sold-out performance generates $9{,}600$ dollars in ticket revenue, how many seats of each type are in the hall?

**7** [Foundational] • [FRQ] • Target: Verifying Solutions to Systems of Inequalities

Determine whether the point $(3, -2)$ lies in the solution region of the system of inequalities $2x - 3y > 10$ and $x + 4y \leq -5$. Justify algebraically.

## Part I — Free-Response Questions (FRQs): Intermediate Modeling (8–14)

**8** [Intermediate] • [FRQ] • Target: Dependent Systems (Infinitely Many Solutions)

Find the values of the constants $a$ and $b$ such that the linear system $ax + 6y = 15$ and $8x - 12y = b$ has infinitely many solutions.

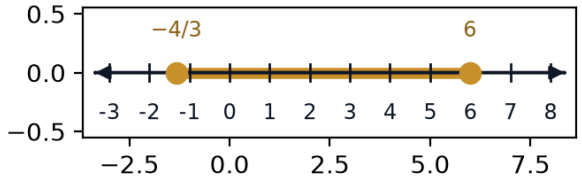
**9** [Intermediate] • [FRQ] • Target: Inconsistent Systems (No Solution)

For what value of $k$ does the linear system $(k - 2)x + 4y = 9$ and $3x + (k + 2)y = 5$ have no solution?

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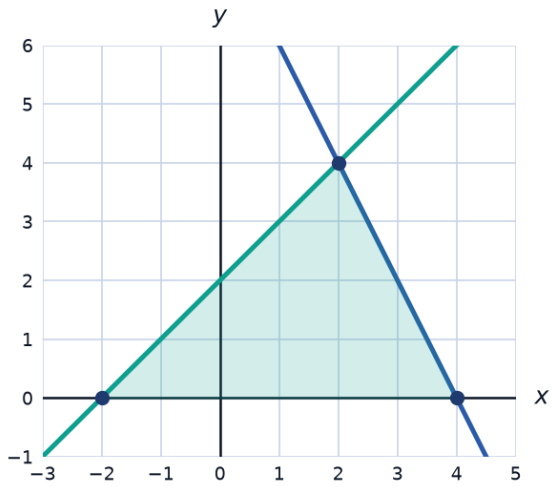
**10** [Intermediate] • [FRQ] • Target: Absolute Value Inequalities & Integer Solutions

Solve the absolute value inequality $|3x - 7| + 4 \le 15$, express the solution set as an interval, and list all integer solutions.




**11** [Intermediate] • [FRQ] • Target: Graphing & Area of Feasible Regions

The shaded region in the $xy$-plane represents the feasible set of the system $y \ge 0$, $y \le x + 2$, and $y \le -2x + 8$. Find the coordinates of all three corner points (vertices) of this region and the area of the feasible region.




**12** [Intermediate] • [FRQ] • Target: Linear Programming Optimization in Context

A coffee roaster blends Ethiopian beans ($14$ dollars/kg) and Colombian beans ($10$ dollars/kg) to make a batch of at least $40$ kg whose total ingredient cost is at most $480$ dollars. Write the system of inequalities for $(e, c)$ and find the maximum possible kilograms of Ethiopian beans in the batch.

**13** [Intermediate] • [FRQ] • Target: Structural Linear Combination Shortcut

Without solving for $x$ and $y$ individually, find the value of $x - y$ given the system $7x + 3y = 41$ and $5x + 5y = 35$.

**14** [Intermediate] • [FRQ] • Target: Relative Velocity Linear Systems

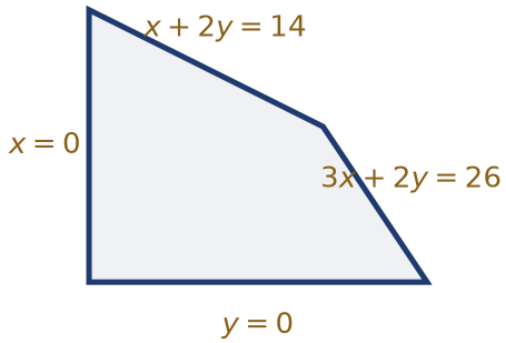
A motorboat travels $48$ km downstream in $1.5$ hours and returns the same $48$ km upstream in $2.4$ hours. Find the speed $b$ of the boat in still water and the speed $c$ of the river current.

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## Part I — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

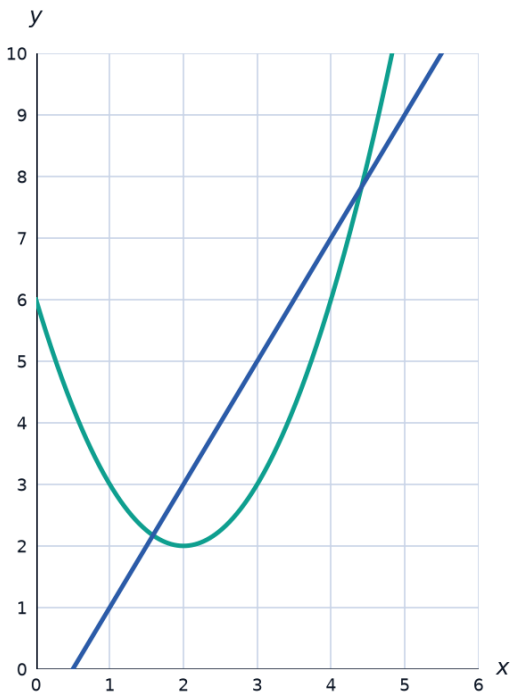
**15** [Challenging] • [FRQ] • Target: Corner-Point Theorem & Linear Programming • Cross-Lesson: Topic 8.01 (Polygonal Regions & Coordinate Geometry)

Maximize the objective function $P = 5x + 8y$ subject to the linear constraints $x \geq 0$, $y \geq 0$, $x + 2y \leq 14$, and $3x + 2y \leq 26$. State the optimal point $(x^*, y^*)$ and the maximum value $P_{\max}$.




**16** [Challenging] • [FRQ] • Target: Linear-Quadratic System & Chord Length • Cross-Lesson: Topic 3.01 (Quadratics) & Topic 7.01 (Distance Formula)

Find all points $(x, y)$ where the line $2x - y = 1$ intersects the quadratic curve $y = x^2 - 4x + 6$, and calculate the exact length of the chord connecting the intersection points.




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**17** [Challenging] • [FRQ] • Target: Piecewise Absolute-Value System Multiplicity • Cross-Lesson: Topic 1.01 (Absolute Value Functions)

For what values of $c$ does the system $y = |x - 2| + 1$ and $y = c$ have (a) zero solutions, (b) exactly one solution, and (c) two distinct solutions?

**18** [Challenging] • [FRQ] • Target: Symmetric Three-Variable Linear Systems • Cross-Lesson: Topic 10.01 (Algebraic Structure & Symmetry)

Solve the three-variable linear system $x + y = 7$, $y + z = 11$, and $z + x = 10$, and compute the product $xyz$.

**19** [Challenging] • [FRQ] • Target: Inscribed Circle of a Linear Feasibility Triangle • Cross-Lesson: Topic 9.01 (Circles & Inradius Formula)

The system of inequalities $x \geq 1$, $y \geq 2$, and $3x + 4y \leq 29$ defines a right triangle in the $xy$-plane. Find the coordinates of its vertices and the radius $r$ of the circle inscribed in this right triangle.

**20** [Challenging] • [FRQ] • Target: Two-Component Percentage Composition System • Cross-Lesson: Topic 5.01 (Percentages & Proportions)

A nutritionist prepares a supplement mix from Powder A ($30\%$ protein, $10\%$ fiber) and Powder B ($15\%$ protein, $25\%$ fiber). To get exactly $180\,\text{g}$ of protein and $110\,\text{g}$ of fiber, how many grams of Powder A and Powder B are required?

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## Part II — Multiple-Choice Questions (MCQs): Foundational Concepts (21–28)

**21** [Foundational] • [MCQ] • Target: Direct Elimination in Linear Systems

If $3x + 2y = 19$ and $x - 2y = 1$, what is the value of $x$?

**a.** 3 **b.** 4

**c.** 5 **d.** 6

**22** [Foundational] • [MCQ] • Target: Sign Reversal in Inequalities

Which of the following is the solution set to the inequality $5 - 3x > 17$?

**a.** $x < -4$ **b.** $x > -4$

**c.** $x < -\frac{22}{3}$ **d.** $x > -\frac{22}{3}$

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**23** [Foundational] • [MCQ] • Target: Solving Linear Systems

If $(x, y)$ is the solution to the system $2x + 5y = 24$ and $3x + 2y = 14$, what is the value of $x + y$?

**a.** 5

**b.** 6

**c.** 7

**d.** 8

**24** [Foundational] • [MCQ] • Target: Translating Verbal Constraints into Inequalities

A bakery sells croissants for $3.50$ dollars each and muffins for $2.25$ dollars each. Maya buys a total of $12$ pastries and spends at most $36.00$ dollars. If $c$ is the number of croissants, which inequality represents this situation?

**a.** $3.50c + 2.25(12 - c) \leq 36.00$

**b.** $3.50c + 2.25(12 - c) \geq 36.00$

**c.** $3.50(12 - c) + 2.25c \leq 36.00$

**d.** $3.50c + 2.25c \leq 36.00$

**25** [Foundational] • [MCQ] • Target: Testing Points in Inequality Systems

Which point $(x, y)$ lies in the solution set of the system $y > 2x - 1$ and $y < -x + 5$?

**a.** $(0, -2)$

**b.** $(1, 2)$

**c.** $(3, 4)$

**d.** $(4, 2)$

**26** [Foundational] • [MCQ] • Target: Collinear Systems

How many solutions $(x, y)$ does the system $4x - 6y = 10$ and $-6x + 9y = -15$ have?

**a.** Zero

**b.** Exactly one

**c.** Exactly two

**d.** Infinitely many

**27** [Foundational] • [MCQ] • Target: Greatest Integer Solutions to Inequalities

What is the greatest integer $x$ that satisfies the inequality $\frac{4x + 3}{5} < 7$?

**a.** 6

**b.** 7

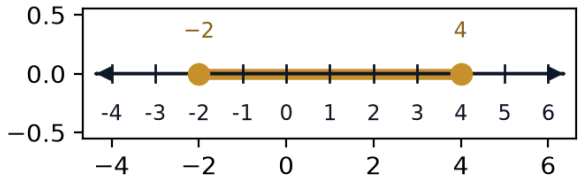
**c.** 8

**d.** 9

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**28** [Foundational] • [MCQ] • Target: Matching Number Lines to Absolute Value Inequalities

The number line below shows the solution set to which inequality?




**a.** $|x - 1| \leq 3$

**b.** $|x + 1| \leq 3$

**c.** $|x - 3| \leq 1$

**d.** $|x - 1| \geq 3$

## Part II — Multiple-Choice Questions (MCQs): Intermediate Exam Practice (29–37)

**29** [Intermediate] • [MCQ] • Target: No-Solution Condition for Linear Systems

In the system of equations $kx - 3y = 8$ and $12x - 9y = 20$, $k$ is a constant. If the system has no solution, what is the value of $k$?

**a.** $-4$

**b.** $3$

**c.** $4$

**d.** $6$

**30** [Intermediate] • [MCQ] • Target: Symmetric Linear Combination

If $7x - 4y = 23$ and $3x + 8y = 5$, what is the value of $5x + 2y$?

**a.** $12$

**b.** $14$

**c.** $16$

**d.** $18$

**31** [Intermediate] • [MCQ] • Target: Two-Constraint Resource Allocation

A landscaping crew can mow a small lawn in $40$ minutes and a large lawn in $75$ minutes. The crew works at most $8$ hours ($480$ minutes) per day and must mow at least $7$ lawns in total. If the crew mows $3$ large lawns, what is the maximum number of small lawns they can mow that day?

**a.** $4$

**b.** $5$

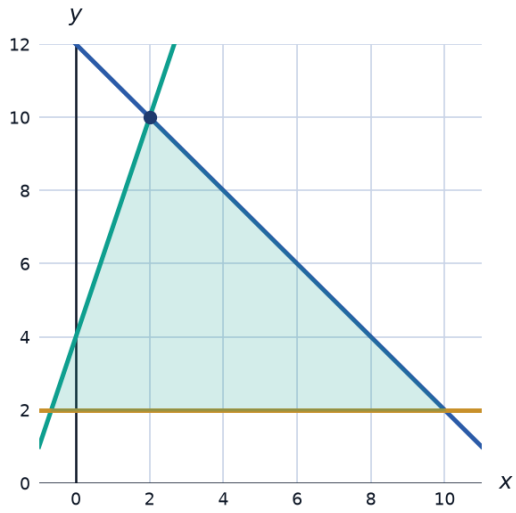
**c.** $6$

**d.** $7$

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**32** [Intermediate] • [MCQ] • Target: Vertex Identification in Linear Inequalities

In the $xy$-plane, the solution set of the system $y \leq 3x + 4$, $y \leq -x + 12$, and $y \geq 2$ is a triangular region. What is the $x$-coordinate of the vertex with the greatest $y$-coordinate?




**a.** 1.5

**b.** 2

**c.** 2.5

**d.** 3

**33** [Intermediate] • [MCQ] • Target: Counting Integer Solutions

For how many integer values of $x$ is the compound inequality $11 < 4x + 3 < 35$ satisfied?

**a.** 4

**b.** 5

**c.** 6

**d.** 7

**34** [Intermediate] • [MCQ] • Target: Intersecting Inequality Constraints

A chemistry lab needs between $300$ mL and $450$ mL of acid mixture. Reagent X contains $20\%$ acid and Reagent Y contains $50\%$ acid. If the lab uses $200$ mL of Reagent X and $y$ mL of Reagent Y, and the total pure acid must be at least $115$ mL, what is the valid range for $y$?

**a.** $100 \leq y \leq 250$

**b.** $120 \leq y \leq 250$

**c.** $150 \leq y \leq 250$

**d.** $150 \leq y \leq 300$

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**35** [Intermediate] • [MCQ] • Target: Rational-Form Linear Systems

If $\frac{x+y}{x-y} = 5$ and $x + y = 10$, what is the value of $xy$?

**a.** 16

**b.** 20

**c.** 24

**d.** 25

**36** [Intermediate] • [MCQ] • Target: Quadrant Analysis of Linear Inequalities

In the $xy$-plane, which quadrants contain NO points $(x, y)$ satisfying the system $y > 2x + 3$ and $y > -\frac{1}{2}x + 1$?

**a.** Quadrant III only

**b.** Quadrant IV only

**c.** Quadrants III and IV

**d.** Quadrants II and IV

**37** [Intermediate] • [MCQ] • Target: System Word Problems for Unit Prices

The cost of $4$ notebooks and $3$ pens is $24.25$ dollars, and the cost of $2$ notebooks and $5$ pens is $18.25$ dollars. What is the cost of $1$ notebook and $1$ pen combined?

**a.** $5.75$ dollars

**b.** $6.25$ dollars

**c.** $6.50$ dollars

**d.** $6.75$ dollars

## Part II — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • [MCQ] • Target: Lattice Points in a Feasible Region • Cross-Lesson: Topic 6.01 (Combinatorics & Counting)

How many ordered pairs of integers $(x, y)$ satisfy the strict inequalities $x > 0$, $y > 0$, and $2x + 3y < 14$?

**a.** 7

**b.** 8

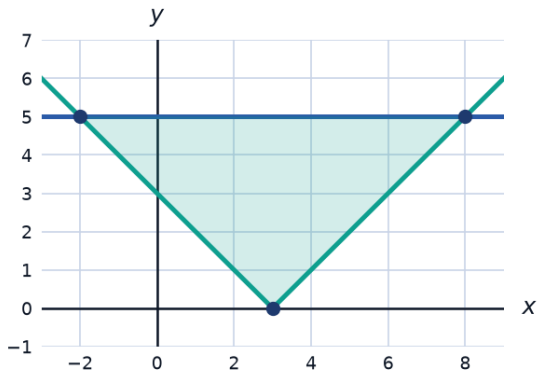
**c.** 9

**d.** 10

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**39** [Challenging] • [MCQ] • Target: Area & Perimeter of Absolute-Value Regions • Cross-Lesson: Topic 7.01 (Isosceles Right Triangles & Special Ratios)

In the $xy$-plane, the system of inequalities $y \geq |x - 3|$ and $y \leq 5$ encloses a triangular region. What is the perimeter of this region?




**a.** $10 + 5\sqrt{2}$

**b.** $10 + 10\sqrt{2}$

**c.** $20$

**d.** $20\sqrt{2}$

**40** [Challenging] • [MCQ] • Target: Determinant Condition for Unique Solutions • Cross-Lesson: Topic 10.01 (Algebraic Structure)

If the linear system $x + ky = 5$ and $kx + 4y = 10$ has a unique solution $(x, y)$, which of the following CANNOT be the value of $k$?

**a.** $-2$ only

**b.** $2$ only

**c.** $0$

**d.** $-2$ and $2$

**41** [Challenging] • [MCQ] • Target: Geometric Probability on a Feasible Region • Cross-Lesson: Topic 6.01 (Probability) & Topic 8.01 (Area)

A point $(x, y)$ is chosen uniformly at random from the rectangle $0 \leq x \leq 6$, $0 \leq y \leq 4$. What is the probability that the point satisfies $x + y \leq 4$?

**a.** $\frac{1}{4}$

**b.** $\frac{1}{3}$

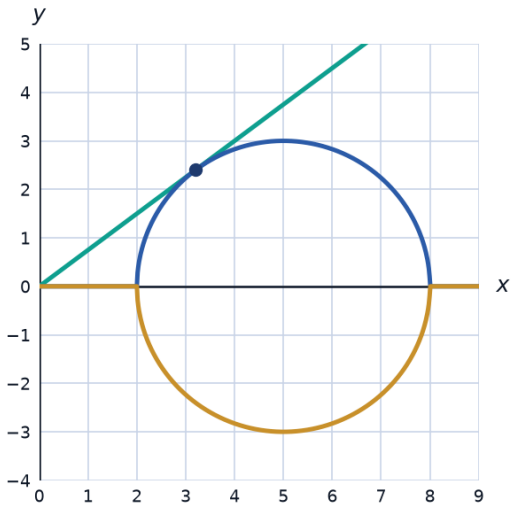
**c.** $\frac{1}{2}$

**d.** $\frac{2}{3}$

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**42** [Challenging] • [MCQ] • Target: Linear-Circle Tangency System • Cross-Lesson: Topic 9.01 (Circles & Tangent Lines)

For what value of $m > 0$ does the system $y = mx$ and $x^2 + y^2 - 10x + 16 = 0$ have exactly one real solution $(x, y)$?




**a.** $\frac{3}{4}$

**b.** $\frac{4}{3}$

**c.** $\frac{3}{5}$

**d.** $\frac{5}{4}$

**43** [Challenging] • [MCQ] • Target: Linear Systems in Transformed Variables • Cross-Lesson: Topic 4.01 (Exponentials & Radicals)

If $a$ and $b$ are positive real numbers satisfying $\log_2(a) + \log_2(b) = 5$ and $3\log_2(a) - \log_2(b) = 7$, what is the value of $\frac{a}{b}$?

**a.** 1

**b.** 2

**c.** 4

**d.** 8

**44** [Challenging] • [MCQ] • Target: Absolute Value Diamond & Half-Plane Area • Cross-Lesson: Topic 8.01 (Area & Coordinate Symmetry)

The region defined by $|x| + |y| \le 6$ in the $xy$-plane is a square. What is the area of the portion of this square that also satisfies $y \ge x - 2$?

**a.** 36

**b.** 48

**c.** 54

**d.** 60

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**45** [Challenging] • [MCQ] • Target: Industrial Production Linear System • Cross-Lesson: Topic 5.01 (Rates & Proportions)

A factory produces two models of solar panels, Model X and Model Y. Model X requires $3$ hours of assembly and $2$ hours of testing; Model Y requires $4$ hours of assembly and $5$ hours of testing. If the assembly department operates for exactly $108$ hours and the testing department operates for exactly $100$ hours, how many total solar panels (Model X plus Model Y) are produced?

**a.** 28

**b.** 30

**c.** 32

**d.** 35

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